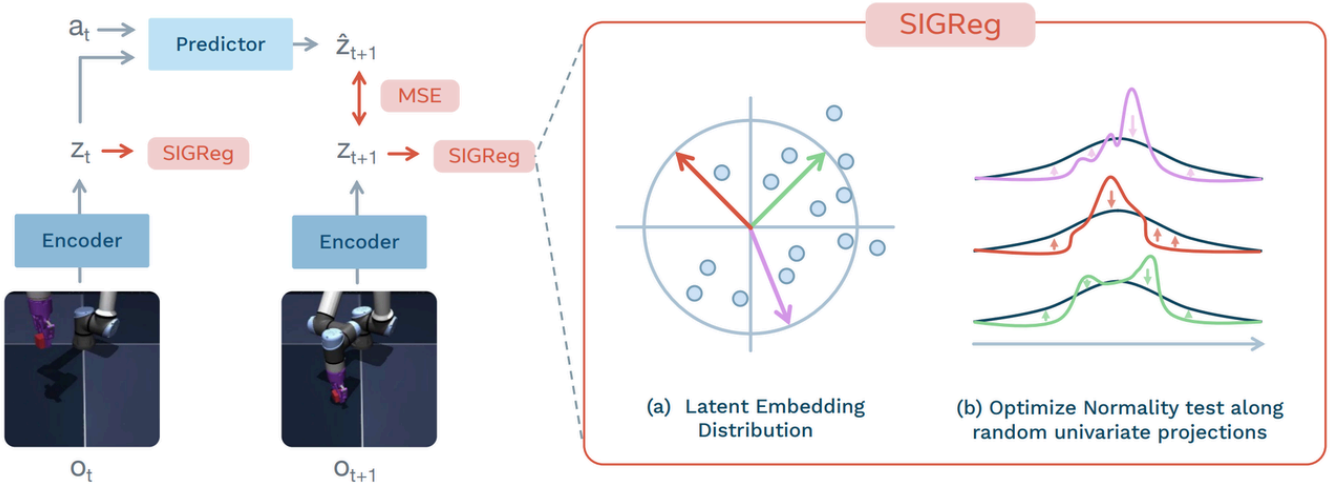
[Le]WorldModel: Stable End-to-End Joint-Embedding Predictive Architecture from Pixels

> #Planning #WorldModel #HybridArchitecture #CV #JEPA



- What problems does this architecture solve?
  - When using the old architecture, JEPAs, as a world model, it is fragile and is easy to cause **representation collapse**.
  - While there are a lot of fix to this problem, these solutions usually have a complex loss function with a set of complicated regularization terms, making the convergence slow and costly.
- How does this architecture work?
  - Architecture:
    - Encoder, enc_θ: A small ViT that will convert the input image into a latent vector z_t.
    - Predictor, pred_ϕ: The world model, a small, 6-layer transformer, that will autoregressively predict the future world state by using the historical context and the current action. $\hat{z}_{t+1} = \mathrm{pred}_\phi(z_t, a_t)$
  - Loss Function

$$\mathcal{L}_{LeWM} \triangleq \underbrace{\mathcal{L}_{pred}}_{\text{Classic Loss}} + \underbrace{\lambda \mathrm{SIGReg}(Z)}_{\text{Regularization for encoder}}$$

    - $\mathcal{L}_{pred}$: The MSE between the predicted future state latent vector and the ground truth future state latent vector that comes from the encoder: $\mathrm{enc}_\theta(o^*_{t+1})$.
    - $\mathrm{SIGReg}(\cdot)$: Sketched-Isotropic-Gaussian Regularizer.
      - To avoid representation collapse, we don't want the latent representation matrix to be a "point" in the space. Instead, we want it to **fill** the space, i.e., to follow an **Isotropic Gaussian** Distribution.
      - However, it is hard to use classic ways to check whether the current distribution has matched the goal distribution in high-dimensional space.
      - **Cramér-Wold Theorem**: If the projections of a high-dimensional distribution onto

all possible one-dimensional directions are exactly equal to the projections of the target distribution onto those same directions, then the two high-dimensional distributions are exactly equal.

- Based on Cramér-Wold Theorem, we can select M directions. For each direction, sample a unit vector: $u^{(m)} \in \mathbb{S}^{d-1}$. Use the unit vector to project the latent matrix in that direction: $h^{(m)} = Zu^{(m)}$. In this way, we only want to check whether each $h^{(m)} \in \mathbb{R}^d$ follows a gaussian distribution.
- We use the **Epps-Pulley** method:

  $$T^{(m)} = \int_{-\infty}^{\infty} w(t)|\phi_N(t; h^{(m)}) - \phi_0(t)|^2 dt \quad \text{where:}$$

  ϕ_N is the empirical characteristic function from $h^{(m)}$

  ϕ_0 is the ECF of the standard GD

  - *If Epps-Pulley is to measure the "similarity/distance" between the current distribution as the penalty of collapse, why don't we use KL divergence? ($h^{(m)}$ and the standard Gaussian distribution)? After all the integral is costly...*
  - Answer from Yann LeCun: $h^{(m)}$ is a discrete distribution and the target distribution is a continuous standard normal distribution. KL cannot be applied to this situation. However, the Fourier-based method above is not perfect as it is expensive and not intuitive.
- Then:

  $\text{SIGReg}(Z) \triangleq \frac{1}{M} \sum_{m=1}^{M} T^{(m)}$
- Why do we use SIGReg to Z? This is because the collapse is from enc_θ. Z is the latent representation of all images in a batch from enc_θ. We want to make sure that the latent representations of these images are different and follow an **Isotropic Gaussian** Distribution.

- Inference:
  - Latent Planning: Use CEM. Sample a series of action from $\mu = 0, \sigma = 1$, compare the final state to the goal state, and optimize μ and σ.
  - Use MPC to avoid accumulation of errors: use the 1st K actions, re-sample the environment, and re-plan.

- Experiment Results
  - Efficient and high-performing: LeWM, with just 15 million parameters, can be trained on a single GPU within a few hours. With the same computational budget, its planning speed is 48 times faster than foundation-model-based world models (such as DINO-WM), while maintaining or even surpassing the latter's success rates on various 2D/3D control tasks (such as PushT and OGBench-Cube).
  - Emergent physical understanding:
    - Implicit encoding of physical quantities: Through linear probing, we demonstrate that the model accurately encodes physical structures such as object coordinates and angles in its latent space.
    - Violation of Expectations (VoE) test: When presented with scenes violating physical continuity—such as "objects teleporting"—the model generates significantly higher prediction errors (Surprise Error), demonstrating that it has learned underlying physical rules from pixels without any prior knowledge. Furthermore, the model exhibits the

naturally emerging phenomenon of "Temporal Straightening" without explicit constraints, where the trajectories of consecutive frames in the representation space gradually smooth out and approach straight lines.

- Work Cited
  - Paper
  - Chat History
  - NotebookLM